

3(a)	<i>one mark for infected person with contaminated faeces</i> faeces / sewage, contaminates, (drinking) water / cooking utensils / vegetable plots / crops / food ; <i>one mark for uninfected person</i> drinking contaminated water / eating food washed in contaminated water ; R infected food or water ; A faecal-oral route <i>for two marks</i>	2
3(b)	<i>any three from:</i> secondary structure ; alpha / α, helix ; <i>bonding</i> hydrogen bonding ; further detail ; e.g. between NH and CO of different amino acids four, amino acids / places, ahead hydrogen of the NH and oxygen of the CO	3
3(c)(i)	<i>any two from:</i> break down / digest / hydrolyse / AW, pathogen / foreign material / AW ; <i>ref. to</i> containing hydrolytic / digestive / named, enzymes ; A hydrolases fuse with the, phagosome / phagocytic vacuole / AW ;	2
3(c)(ii)	<i>any two from:</i> cells are dead so cannot, cause disease / reproduce / AW ; only part of a cholera toxin molecule, so no toxic effect ; recognised by / trigger clonal selection of, more than one type of B-lymphocyte ; stimulates production of different antibodies ; stimulates the production of more than one type of memory cell / AW ; <i>ref. to</i> immune system cells can respond to bacterial cells and the toxin produced ; AVP ; e.g. different antigens on the surface of the cells	2

Question	Answer	Marks
3(d)	any four from: (resistant) (<i>Vibrio cholerae</i>) is not killed by the antibiotic ; A can still reproduce inability to treat people with cholera using antibiotics ; prolonged periods of illness ; (so) increased risk of spread / AW ; increased death rate from cholera ; transfer of antibiotic resistance (alleles) to, other species of bacteria / other strains of <i>V.cholerae</i> ; development of multiple resistance in other species of bacteria ; inability to treat other bacterial diseases ; increased death rate from other bacterial diseases ; need to research alternative antibiotics ; AVP ;	4

Question	Answer	Marks
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